STRATOSYN / LIFE SCIENCE

Cumulative Final

Grade 7 · Rm 501

FINAL EXAM · 7TH GRADE LIFE SCIENCE

# The Whole

80 questions covering every unit from this year — non-contact forces, cells, human body systems, extreme weather and biomes, plate tectonics, and the periodic table. Q1-65 multiple choice, Q66-75 fill-in-the-blank, Q76-80 free response. Click an option to select it (you can change your mind). When you are finished, fill in your name at the top, then hit Copy My Answers at the bottom to grab everything formatted for Canvas, email, or printing.

MS. MADEWELL · ROOM 501 · ALA VISTANCIA

65 Multiple Choice | 10 Fill-in | 5 Free Response | Submit by Copy

## SECTION A · MULTIPLE CHOICE (1-65)

### Question 1

NON-CONTACT FORCES

Which of the following is a **non-contact** force?

A Friction

B Tension in a rope

C Gravity

D A push from your hand

## Question 2

NON-CONTACT FORCES

Two magnets are placed near each other and they push apart. This means:

A Like poles are facing each other (N-N or S-S)

B Opposite poles are facing each other

C One of the magnets is broken

D The magnets have no charge

## Question 3

NON-CONTACT FORCES

As the distance between two objects with mass **increases**, the gravitational force between them:

A Stays exactly the same

B Increases

C Decreases

D Becomes negative

## Question 4

NON-CONTACT FORCES

A balloon is rubbed on someone's hair, then sticks to a wall. The non-contact force responsible is:

A Magnetism

B Gravity

C Static electricity

D Friction

## Question 5

NON-CONTACT FORCES

When an object becomes **negatively** charged, what particles have moved onto it?

A Protons

B Neutrons

C Electrons

D Whole atoms

## Question 6

NON-CONTACT FORCES

Gravity on the Moon is about 1/6 of Earth's. An astronaut's **mass** on the Moon, compared to Earth, is:

A 1/6 as much

B 6 times greater

C Exactly the same

D Zero

## Question 7

NON-CONTACT FORCES

The region of space around a magnet where it can exert a force on another magnet is called the:

A Magnetic field

B Charge cloud

C Gravity well

D Static zone

## Question 8

NON-CONTACT FORCES

Which best describes **weight**?

A The amount of matter in an object

B The pull of gravity on an object's mass

C The volume of an object

D The speed at which an object falls

## Question 9

NON-CONTACT FORCES

Object A has a charge of **+2**. Object B has a charge of **−3**. The electric force between them will be:

A Repulsive (push apart)

B Attractive (pull together)

C Zero

D Magnetic only

## Question 10

NON-CONTACT FORCES

If you snap a bar magnet exactly in half, what happens?

A You get one piece with N pole and one piece with S pole, separated

B Both pieces lose their magnetism

C You get two new magnets, each with its own N pole and S pole

D One piece keeps both poles, the other has none

## Question 11

CELLS

The basic unit of all living things is the:

A Atom

B Tissue

C Cell

D Organ

## Question 12

CELLS

Which organelle is often called the **"powerhouse of the cell"** because it produces most of the cell's energy?

A Nucleus

B Mitochondria

C Ribosome

D Vacuole

## Question 13

CELLS

Which structure is found in **plant** cells but NOT in animal cells?

A Nucleus

B Mitochondria

C Cell wall

D Ribosome

## Question 14

CELLS

Photosynthesis takes place primarily in which organelle?

A Mitochondria

B Nucleus

C Chloroplast

D Lysosome

## Question 15

CELLS

Which statement is **NOT** part of the cell theory?

A All living things are made of one or more cells

B The cell is the basic unit of life

C All cells come from other cells

D Every cell must contain a nucleus

## Question 16

CELLS

The **"control center"** of a eukaryotic cell, where DNA is stored, is the:

A Mitochondria

B Nucleus

C Vacuole

D Cell membrane

## Question 17

CELLS

Prokaryotic cells (such as bacteria) do **NOT** have which of the following?

A DNA

B A cell membrane

C Cytoplasm

D A membrane-bound nucleus

## Question 18

CELLS

The thin barrier that controls what enters and leaves a cell is the:

A Cell wall

B Nucleus

C Cell membrane

D Vacuole

## Question 19

CELLS

Which list shows the levels of organization from **smallest to largest**?

A Cell → tissue → organ → organ system

B Tissue → cell → organ → organ system

C Organ → tissue → cell → organ system

D Organism → organ → tissue → cell

## Question 20

CELLS

Which equation best summarizes **photosynthesis**?

A Glucose + $O_2$ → $CO_2$ + $H_2O$ + energy

B $CO_2$ + $H_2O$ + light → glucose + $O_2$

C Water → hydrogen + oxygen

D ATP → ADP + heat

## Question 21

CELLS

Cellular respiration releases usable energy by breaking down which molecule?

A Glucose

B DNA

C Water

D Cell walls

## Question 22

HUMAN BODY SYSTEMS

Which body system pumps blood through the body?

A Respiratory

B Nervous

C Circulatory

D Digestive

## Question 23

HUMAN BODY SYSTEMS

The **diaphragm** is a major muscle for which body system?

A Skeletal

B Respiratory

C Digestive

D Excretory

## Question 24

HUMAN BODY SYSTEMS

Where does most **chemical digestion** and absorption of nutrients occur?

A Stomach

B Esophagus

C Small intestine

D Large intestine

## Question 25

HUMAN BODY SYSTEMS

Which system is made up of the brain, spinal cord, and nerves and sends signals throughout the body?

A Endocrine

B Nervous

C Skeletal

D Muscular

## Question 26

HUMAN BODY SYSTEMS

The main job of **red blood cells** is to carry:

A Oxygen

B Hormones only

C Solid wastes

D Glucose only

## Question 27

HUMAN BODY SYSTEMS

Bones meet other bones at structures called:

A Tissues

B Organs

C Joints

D Tendons

## Question 28

HUMAN BODY SYSTEMS

The kidneys, which filter wastes from the blood, are part of which system?

A Digestive

B Excretory

C Respiratory

D Endocrine

## Question 29

HUMAN BODY SYSTEMS

**Hormones** are chemical messengers released by which system?

A Nervous

B Endocrine

C Skeletal

D Integumentary

## Question 30

HUMAN BODY SYSTEMS

The respiratory and circulatory systems work **together** to:

A Digest food into nutrients

B Exchange oxygen and carbon dioxide between air and blood

C Filter and remove urine

D Send electrical signals to the brain

## Question 31

HUMAN BODY SYSTEMS

Skin, hair, and nails make up the:

A Integumentary system

B Endocrine system

C Skeletal system

D Lymphatic system

## Question 32

HUMAN BODY SYSTEMS

Which is **NOT** a function of the skeletal system?

A Supports the body

B Protects organs like the brain and lungs

C Produces blood cells in bone marrow

D Pumps blood

## Question 33

WEATHER & BIOMES

What is the difference between **weather** and **climate**?

A Climate is short-term, weather is long-term

B Weather is day-to-day; climate is the long-term average over many years

C They mean exactly the same thing

D Climate only describes deserts

## Question 34

WEATHER & BIOMES

A rapidly rotating column of air that touches the ground is a:

A Hurricane

B Blizzard

C Tornado

D Drought

## Question 35

WEATHER & BIOMES

Hurricanes form over:

A Cold ocean water

B Warm ocean water near the equator

C Mountain ranges

D Polar ice sheets

## Question 36

WEATHER & BIOMES

A long period of below-average rainfall is called a:

A Hurricane

B Drought

C Flood

D Tornado

## Question 37

WEATHER & BIOMES

Which biome has the **highest biodiversity** (the most different species)?

A Tundra

B Desert

C Tropical rainforest

D Taiga

## Question 38

WEATHER & BIOMES

The biome characterized by **permafrost** (permanently frozen soil) and short summers is the:

A Tundra

B Tropical rainforest

C Grassland

D Desert

## Question 39

WEATHER & BIOMES

Cacti, kangaroo rats, and rattlesnakes are all adapted to which biome?

A Tundra

B Taiga

C Desert

D Marine

## Question 40

WEATHER & BIOMES

The **vast majority** of Earth's water is found in the:

A Lakes and rivers

B Glaciers and ice caps

C Oceans

D Atmosphere

## Question 41

WEATHER & BIOMES

The amount of **solar energy** a region receives depends most on its:

A Time of day only

B Latitude (distance from the equator)

C Wind speed

D Plant cover

## Question 42

WEATHER & BIOMES

In the **eye** of a hurricane, air pressure is:

A Higher than the surrounding air

B Equal to the surrounding air

C Lower than the surrounding air

D Always exactly zero

## Question 43

WEATHER & BIOMES

Which biome typically has **4 distinct seasons** and trees that lose their leaves each fall (deciduous)?

A Tropical rainforest

B Temperate forest

C Tundra

D Desert

## Question 44

PLATE TECTONICS

The **thinnest, outermost** layer of Earth (where we live) is the:

A Mantle

B Outer core

C Inner core

D Crust

## Question 45

PLATE TECTONICS

Which layer of Earth is a **liquid** made mostly of melted iron and nickel?

A Inner core

B Outer core

C Crust

D Lithosphere

## Question 46

PLATE TECTONICS

The scientist who first proposed that the continents had drifted apart over time was:

A Isaac Newton

B Alfred Wegener

C Charles Darwin

D James Hutton

## Question 47

PLATE TECTONICS

The supercontinent that existed about 250 million years ago, before breaking apart, was:

A Gondwana

B Laurasia

C Pangaea

D Atlantis

## Question 48

PLATE TECTONICS

Tectonic plates move because of:

A Earth's rotation alone

B The pull of the magnetic poles

C Convection currents in the mantle

D Ocean tides on the surface

## Question 49

PLATE TECTONICS

At a **divergent** plate boundary, plates are:

A Sliding past each other

B Moving apart from each other

C Colliding head-on

D Completely still

## Question 50

PLATE TECTONICS

A **mid-ocean ridge** forms at which type of plate boundary?

A Convergent (continent-continent)

B Divergent (oceanic)

C Transform

D No boundary — it's a hot spot

## Question 51

PLATE TECTONICS

The **Himalayan mountains** formed when two continental plates collided. This is what type of boundary?

A Divergent

B Transform

C Convergent (continent-continent)

D Hot spot

## Question 52

PLATE TECTONICS

The **San Andreas Fault** in California is a famous example of what kind of plate boundary?

A Convergent boundary

B Divergent boundary

C Transform boundary

D Subduction zone

## Question 53

PLATE TECTONICS

When a denser **oceanic plate** meets a less dense **continental plate**, the oceanic plate:

A Slides on top of the continent

B Subducts (sinks below) the continental plate

C Pushes the continent straight up

D Stops moving completely

## Question 54

PLATE TECTONICS

The rigid outer shell of Earth (the crust + upper mantle), broken into tectonic plates, is called the:

A Asthenosphere

B Lithosphere

C Hydrosphere

D Mesosphere

## Question 55

PERIODIC TABLE

The **atomic number** of an element equals the number of:

A Neutrons

B Electrons in the outermost shell

C Protons

D Atomic mass in grams

## Question 56

PERIODIC TABLE

The **mass number** of an atom equals:

A Protons + electrons

B Protons + neutrons

C Just the neutrons

D Protons × electrons

## Question 57

PERIODIC TABLE

The horizontal **rows** on the periodic table are called:

A Groups

B Families

C Periods

D Series

## Question 58

PERIODIC TABLE

The vertical **columns** on the periodic table are called:

A Periods

B Groups (or families)

C Rows

D Levels

## Question 59

PERIODIC TABLE

The element with the symbol **Na** is:

A Nitrogen

B Sodium

C Neon

D Nickel

## Question 60

PERIODIC TABLE

The **noble gases** (Group 18) are best described as:

A Highly reactive

B Inert (very unreactive)

C Magnetic metals

D Liquid at room temperature

## Question 61

PERIODIC TABLE

Which group contains the most **reactive metals**?

A Group 1 (alkali metals)

B Group 18 (noble gases)

C Group 14

D Group 17 (halogens)

## Question 62

PERIODIC TABLE

Which subatomic particle has **no electric charge**?

A Proton

B Electron

C Neutron

D Photon

## Question 63

PERIODIC TABLE

Which element is a **metalloid** (has properties of both metals and nonmetals)?

A Iron (Fe)

B Silicon (Si)

C Sodium (Na)

D Oxygen (O)

## Question 64

PERIODIC TABLE

An atom of **carbon (C)** has atomic number 6. How many **protons** does it have?

A 0

B 6

C 12

D 14

## Question 65

PERIODIC TABLE

Most of the elements on the periodic table are classified as:

A Nonmetals

B Noble gases

C Metals

D Metalloids

SECTION B · FILL-IN-THE-BLANK (66-75)

## Question 66

FILL-IN · CELLS

The organelle that acts as the **"control center"** of a eukaryotic cell and contains the cell's DNA is the _____.

Your answer...

## Question 67

FILL-IN · PERIODIC TABLE

The chemical **symbol** for the element sodium is _____.

Your answer...

## Question 68

FILL-IN · PLATE TECTONICS

The supercontinent that existed about 250 million years ago, before the modern continents drifted to where they are today, was called _____.

Your answer...

## Question 69

FILL-IN · PERIODIC TABLE

An atom of oxygen has an atomic number of 8. The number of **protons** in the nucleus of this atom is _____.

Your answer...

## Question 70

FILL-IN · BIOMES

A cold biome with permafrost (permanently frozen ground), low precipitation, and short summers, found near the Arctic Circle, is called the _____.

Your answer...

## Question 71

FILL-IN · PLATE TECTONICS

The type of plate boundary at which two tectonic plates pull **away** from each other is called a _____ boundary.

Your answer...

## Question 72

FILL-IN · BODY SYSTEMS

The body system that pumps blood through arteries, veins, and capillaries to carry oxygen, nutrients, and wastes is called the _____ system.

Your answer...

## Question 73

FILL-IN · NON-CONTACT FORCES

The non-contact force that pulls every object with mass toward every other object with mass — including what holds the Moon in orbit around Earth — is _____.

Your answer...

## Question 74

FILL-IN · CELLS

The process by which plants use sunlight, water, and carbon dioxide to make their own food (glucose) is called _____.

Your answer...

## Question 75

FILL-IN · PLATE TECTONICS

Wegener's idea that the continents have slowly moved over Earth's surface throughout history is called the theory of _____ _____ (two words).

Your answer...

SECTION C · FREE RESPONSE (76-80)

## Question 76

FRQ · CELLS

**Compare and contrast** a plant cell and an animal cell.

(a) List at least **three** organelles found in BOTH plant and animal cells, and briefly describe the job of each one.

(b) List at least **two** structures found ONLY in plant cells and explain why a plant needs each one.

(c) Animal cells often look round or irregular while plant cells look more like rectangles or boxes. Explain *why* plant cells have this shape.

*Write your full response here. Use complete sentences and label your parts (a), (b), (c).*

## Question 77

FRQ · BODY SYSTEMS

Describe how the **respiratory system** and the **circulatory system** work TOGETHER to deliver oxygen from the air all the way to the cells in your body, and to remove carbon dioxide.

In your answer, you must use and correctly explain at least **five** of these terms: lungs, alveoli, diaphragm, red blood cells, hemoglobin, capillaries, heart, oxygen, carbon dioxide, mitochondria.

Step-by-step is fine — pretend you are tracing one molecule of oxygen from the air to a cell.

*Walk through the path of an oxygen molecule and a carbon dioxide molecule. Use the listed vocabulary.*

## Question 78

FRQ · PLATE TECTONICS

Compare the three main types of **plate boundaries**: convergent, divergent, and transform.

For **each** of the three boundary types, your answer must include:
**(a)** what the plates are doing (their motion),
**(b)** one geological feature or event that forms there (e.g., a mountain range, mid-ocean ridge, volcano, earthquake), and
**(c)** one real-world example (a place on Earth).

*Organize your answer by boundary type. You can use bullet points or paragraphs.*

## Question 79

FRQ · WEATHER & BIOMES

The **tropical rainforest** biome has very high biodiversity, while the **desert** biome has much lower biodiversity.

(a) Identify at least **three abiotic factors** (non-living parts of the environment) that explain why the rainforest can support so many species.

(b) Identify at least **two abiotic factors** that make the desert a much harsher environment for life.

(c) Describe two **adaptations** — one of an organism in the rainforest, and one of an organism in the desert — and explain how each adaptation helps that organism survive.

*Be specific. Abiotic factors include things like sunlight, temperature, water, soil, etc.*

## Question 80

FRQ · PERIODIC TABLE + FORCES

An atom of **magnesium (Mg)** has an atomic number of **12** and a mass number of **24**.

(a) How many **protons**, **neutrons**, and **electrons** does a ***neutral*** magnesium atom have? Show how you figured each one out.

(b) In which **group** and **period** of the periodic table is magnesium located? (You may use a periodic table.)

(c) Magnesium is a metal. List **two** general physical properties of metals.

(d) Two magnesium ions, each with a charge of **+2**, are placed near each other. What kind of **non-contact force** exists between them, and is the force **attractive** or **repulsive**? Explain why, using what you know about charges.

*Label parts (a), (b), (c), (d). Show your reasoning for (a) and (d).*

## Submit Your Answers

Click below to copy a clean, formatted version of all your answers. Paste it into Canvas, email, or anywhere else Ms. Madewell asks you to submit.

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